

HONGWEI WU

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EDUCATION

Purdue University Department of Computer Science	West Lafayette, IN
<i>Ph.D. in Computer Science</i>	<i>August 2020 - May 2027 (Expected)</i>
Binghamton University, SUNY Watson College of Engineering	Binghamton, NY
<i>M.S. in Computer Science (Distinguish Graduate Student)</i>	<i>August 2018 - May 2020</i>
Renmin University of China School of Information	Beijing, China
<i>B.S. in Information Security</i>	<i>September 2015 - June 2019</i>

TECHNICAL SKILLS

- Expertise: Program Analysis, Binary Analysis, Reverse Engineering, Decompiler, Embedded System, Large Language Model, Fuzzing
- Languages: Python, Assembly, C/C++, LLVM-IR
- Tools: angr (Core Contributor), IDA Pro, Ghidra, Joern, LLVM, GDB, Intel Pin

RESEARCH EXPERIENCE

Purdue University	West Lafayette, IN
<i>Research Assistant advised by Dr. Antonio Bianchi</i>	<i>August 2020 - Present</i>
• Developed <i>Patch2Text</i>, an LLM-based pipeline generating human-readable descriptions of binary patches from stripped binaries by pairing denoising (stripping renames and address shifts from compilation/decompilation) and a multi-layer diff refinement with an agentic LLM generator; reached 0.92 F1 on a 500-commit benchmark, a +0.242 F1 gain over the raw-decompiled-diff baseline. • Developed <i>VeriBin</i>, an adaptive system designed to verify patch safety without requiring source code by employing symbolic execution to detect patch-introduced modifications, adaptively query analysts for these modifications, and verify whether the patch preserves the original functionality. Achieved 93% accuracy with no false positives on a dataset of 86 samples. • Applied and expanded <i>VeriBin</i> in DARPA's Assured Micropatching Program to verify binary-level micro-patches for legacy binary systems without access to the source code (Cummins engine ECUs, NASA Lunar rovers, and power grid infrastructures, etc.), by enhancing the symbolic execution of <i>angr</i> for various embedded architectures (ARM, PowerPC, etc.) and collaborating with industry partners to integrate the tool into strategic frameworks.	

- Spearheaded the development of *Artiphishell* for DARPA’s AI Cybersecurity Challenge (AIxCC) with team Shellphish, creating an LLM-based Cyber Reasoning System that autonomously identifies and patches vulnerabilities; won a \$2M semifinal award and placed 5th in the final competition at DEF CON 33 (out of 42 initial teams) by uniquely patching real-world vulnerabilities and validating LLM-generated fixes to ensure functional safety.

Binghamton University

Binghamton, NY

Research Assistant advised by Dr. Aravind Prakash

June 2019 - May 2020

- Evaluated the effectiveness of debloating methods by implementing Intel Pin-based Pintools to measure runtime instruction execution and utilizing WinAFL to analyze code coverage of WinRAR using debloated libraries.
- Reproduced rowhammer attacks on machines with non-ECC DRAM to engender bit flips and designed an algorithm to generate a unique identifier for victim machines based on their unique bit-flipping patterns under rowhammer attack.

PUBLICATIONS

- **Hongwei Wu**, Jianliang Wu, Ruoyu Wu, Ayushi Sharma, Aravind Machiry, and Antonio Bianchi, “VeriBin: Adaptive Verification of Patches at the Binary Level” In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2025.
- **Hongwei Wu**, Han Dai, Antonio Bianchi, “Patch2Text: Generating Human Readable Descriptions of Binary Patches from Stripped Binaries”, (*In submission to ACM SIGSAC Conference on Computer and Communications Security (CCS)*), 2026.
- Muqi Zou, Hongyu Cai, **Hongwei Wu**, Zion Leonahenahe Basque, Arslan Khan, Berkay Celik, Dave Tian, Antonio Bianchi, Ruoyu Wang, and Dongyan Xu, “D-LiFT: Improving LLM-based Decompiler Backend via Code Quality-driven Fine-tuning”, (*In submission*), 2026.